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Lab:2

Basic Syntax and Input / Output in C++

Objective of the Lab

The objective of this lab is to understand the **basic syntax of C++ programming language**, including **data types, indentation, comments, and arithmetic operators**. The lab also helps students learn how to use **input and output statements** and perform simple calculations using C++.

Introduction

C++ is a powerful programming language used to develop software, applications, and systems. In this lab we learned the **basic structure of a C++ program**, how to declare variables of different data types, how to take input from the user, and how to display results on the screen using **cout and cin**.

C++ is a widely used programming language that is used to build many types of software applications. Before writing complex programs, it is important to understand the **basic structure and syntax of the language**.

Every C++ program follows a specific structure that includes:

- Header files

- The main function
- Variable declarations
- Statements and expressions
- Output statements

In this lab, we practiced declaring variables of different **data types**, displaying their **memory size and memory location**, and writing simple programs that perform **mathematical calculations**.

We also learned about **indentation and comments**, which help make the code more readable and understandable for programmers.

Task 1

Declare different types of variables and find their memory size, location, and values.

Program

```
#include <iostream>
using namespace std;

int main()
{
    int a = 10;
    float b = 5.5;
    char c = 'A';
    double d = 15.25;

    cout << "Integer value: " << a << endl;
    cout << "Memory size of int: " << sizeof(a) << " bytes" << endl;
    cout << "Location of int: " << &a << endl << endl;

    cout << "Float value: " << b << endl;
    cout << "Memory size of float: " << sizeof(b) << " bytes" << endl;
    cout << "Location of float: " << &b << endl << endl;

    cout << "Char value: " << c << endl;
    cout << "Memory size of char: " << sizeof(c) << " bytes" << endl;
    cout << "Location of char: " << &c << endl << endl;

    cout << "Double value: " << d << endl;
    cout << "Memory size of double: " << sizeof(d) << " bytes" << endl;
    cout << "Location of double: " << &d << endl;

    return 0;
}
```

Explanation

- Different variables (**int, float, char, double**) are declared.

- **sizeof()** is used to find the memory size.
- **& operator** is used to display the memory location (address) of variables.
- **cout** displays the values on the screen.

Task 2

Declare and display a floating-point variable called “pi”.

Program

```
#include <iostream>
using namespace std;

int main()
{
    float pi = 3.14;

    cout << "Value of pi is: " << pi << endl;

    return 0;
}
```

Explanation

- A floating-point variable **pi** is declared.
- The value **3.14** is assigned to it.
- The value is displayed using **cout**.

Task 3

Convert temperature from Celsius to Fahrenheit.

Formula

F=(C×9/5)+32F = (C \times 9/5) + 32F=(C×9/5)+32

Program

```
#include <iostream>
using namespace std;

int main()
{
    float celsius, fahrenheit;

    cout << "Enter temperature in Celsius: ";
```

```
    cin >> celsius;

    fahrenheit = (celsius * 9/5) + 32;

    cout << "Temperature in Fahrenheit is: " << fahrenheit << endl;

    return 0;
}
```

Explanation

- The program takes **Celsius temperature as input** from the user.
- It converts the value using the **formula**.
- The result is displayed using **cout**.

In this lab we learned the **basic syntax of C++**, including **data types, comments, indentation, and arithmetic operations**. We also practiced declaring variables, finding their memory size and location, and writing simple programs for displaying values and performing temperature conversion.